

Duration: **110 minutes**  
Aids Allowed: **NONE**

Student Number: \_\_\_\_\_

Last (Family) Name: \_\_\_\_\_

First (Given) Name(s): \_\_\_\_\_

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*Do **not** turn this page until you have received the signal to start.*  
(In the meantime, please fill out the identification section above,  
and read the instructions below *carefully*.)

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This term test consists of 6 questions on 9 pages (including this one), *When you receive the signal to start, please make sure that your copy of the test is complete.*

Answer each question directly on the test paper, in the space provided, and use the reverse side of the last page for rough work. If you need more space for one of your solutions, use the reverse side of the final page and *indicate clearly the part of your work that should be marked.*

**General Hints:** We were careful to leave ample space on the test paper to answer each question. Note that the exam is worth 60 marks, so if you average more than 2min per mark, you may run out of time.

#### MARKING GUIDE

# 1: \_\_\_\_\_/10

# 2: \_\_\_\_\_/10

# 3: \_\_\_\_\_/10

# 4: \_\_\_\_\_/10

# 5: \_\_\_\_\_/10

# 6: \_\_\_\_\_/10

TOTAL: \_\_\_\_\_/60

*Good Luck!*

**Question 1.** [10 MARKS]

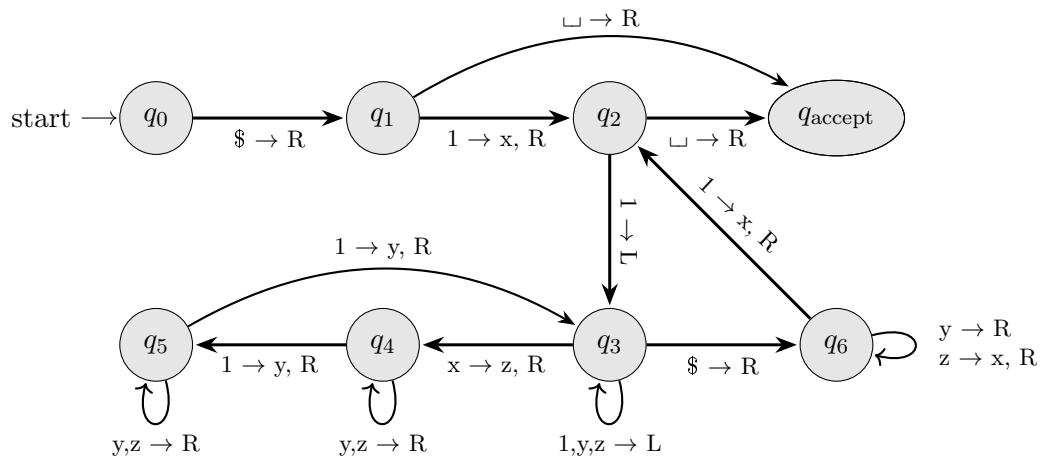
Consider the following Turing machine:

|                                                         |               |                |              |       |            |
|---------------------------------------------------------|---------------|----------------|--------------|-------|------------|
| States: $q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_{accept}$ |               |                |              |       |            |
| Input alphabet: $\{\$, 0, 1\}$                          |               |                |              |       |            |
| Tape alphabet: $\{\$, 0, 1, -, x, y, z\}$               |               |                |              |       |            |
|                                                         | current state | character read | next state   | write | left/right |
|                                                         | $q_0$         | $\$$           | $q_2$        | none  | R          |
|                                                         | $q_1$         | 1              | $q_2$        | $x$   | R          |
|                                                         | $q_1$         | -              | $q_{accept}$ | none  | R          |
|                                                         | $q_2$         | 1              | $q_3$        | none  | L          |
|                                                         | $q_3$         | 1, $y, z$      | $q_3$        | none  | L          |
|                                                         | $q_3$         | $x$            | $q_4$        | $z$   | R          |
|                                                         | $q_3$         | $\$$           | $q_6$        | none  | R          |
| Arrows:                                                 | $q_4$         | $y, z$         | $q_4$        | none  | R          |
|                                                         | $q_4$         | 1              | $q_5$        | $y$   | R          |
|                                                         | $q_5$         | $y, z$         | $q_5$        | none  | R          |
|                                                         | $q_5$         | 1              | $q_3$        | $y$   | R          |
|                                                         | $q_5$         | 0              | $q_6$        | $x$   | L          |
|                                                         | $q_6$         | $y$            | $q_6$        | none  | R          |
|                                                         | $q_6$         | $z$            | $q_6$        | $x$   | R          |
|                                                         | $q_6$         | 1              | $q_2$        | $x$   | R          |

Start at  $q_0$ .

Accept state:  $q_{accept}$

The reject state is implicit.



If this machine is run on the input  $\$1111\$$ , its first configuration will be  $q_0\$1111\$$ . What will the remaining configurations be? Fill in the chart below.

**Solution:**

|                                         |                                         |                                            |                                            |
|-----------------------------------------|-----------------------------------------|--------------------------------------------|--------------------------------------------|
| $C_1: \$\mathbf{q_1 1 1 1 1} \$ \sqcup$ | $C_5: \$z y q_5 1 1 \$ \sqcup$          | $C_9: \$\mathbf{q_3 z y y 1} \$ \sqcup$    | $C_{13}: \$x y q_6 y 1 \$ \sqcup$          |
| $C_2: \$x q_2 1 1 1 \$ \sqcup$          | $C_6: \$z y y q_3 1 \$ \sqcup$          | $C_{10}: q_3 \$z y y 1 \$ \sqcup$          | $C_{14}: \$\mathbf{x y y q_6 1} \$ \sqcup$ |
| $C_3: \$q_3 x 1 1 1 \$ \sqcup$          | $C_7: \$z y q_3 y 1 \$ \sqcup$          | $C_{11}: \$\mathbf{q_6 z y y 1} \$ \sqcup$ | $C_{15}: \$x y y x q_2 \$ \sqcup$          |
| $C_4: \$\mathbf{z q_4 1 1 1} \$ \sqcup$ | $C_8: \$\mathbf{z q_3 y y 1} \$ \sqcup$ | $C_{12}: \$x q_6 y y 1 \$ \sqcup$          | $C_{16}: \$x y y x \$ q_R \sqcup$          |

**Question 2.** [10 MARKS]

Suppose that you have three languages:  $A$ ,  $B$ , and  $C$ . You know that  $A \leq_m B$  and  $B \leq_m C$ . Prove that  $A \leq_m C$ .

**Solution:**

Since  $A \leq_m B$ , we know that there is some computable  $f$  such that  $x \in A$  iff  $f(x) \in B$ . (\*)

Similarly, since  $B \leq_m C$ , we know that there is some computable  $g$  such that  $y \in B$  iff  $g(y) \in C$ . (\*\*)

Putting these together, we get this reduction:

Let  $P =$  “On input  $\langle x \rangle$ :

1. Let  $y = f(x)$ .
2. Let  $z = g(y)$ .
3. Return  $\langle z \rangle$ .”

Since both  $f$  and  $g$  both halt on every input, so does  $P$ .

**Proof that  $\langle x \rangle \in A$  iff  $\langle z \rangle \in C$ :**

$\Rightarrow$ ) Suppose that  $\langle x \rangle \in A$ .

Then,  $f(x) = y$  is in  $B$ , by (\*).

So  $g(y) = z$  is in  $C$ , by (\*\*).

$\Leftarrow$ ) Suppose that  $\langle x \rangle \notin A$ .

Then,  $f(x) = y$  is not in  $B$ , by (\*).

So  $g(y) = z$  is not in  $C$ , by (\*\*).

**Question 3.** [10 MARKS]

Consider the language  $L_3$ :

$$L_3 = \left\{ \langle M \rangle \mid \begin{array}{l} M \text{ is a TM, and if } M \text{ accepts any string,} \\ \text{then } M \text{ loops on some (other) string with the same length} \end{array} \right\}.$$

Prove that  $L_3$  is co-recognizable.

**Solution:**

Firstly, here's the definition of  $\overline{L_3}$ :

$$\overline{L_3} = \left\{ \langle M \rangle \mid \begin{array}{l} M \text{ is a TM, and } \exists \text{ a string } x \text{ such that } M \text{ accepts } x, \\ \text{and } M \text{ halts on every string with the same length} \end{array} \right\}$$

The certificate will be of the form  $\langle x, k \rangle$ , where  $x$  is an input that  $M$  accepts, and where  $k$  is the maximum number of steps you'll need to run  $M$  to get it to halt on any input of size  $|x|$ .

Finally, Here's a recognizer for  $\overline{L_3}$ :

Let  $w_1, w_2, \dots$  be an effective enumeration of  $\Sigma^*$ .

$R =$  "On input  $\langle M \rangle$ :

1. For  $k = 1$  to  $\infty$ :
2.   For  $j = 1$  to  $k$ :"
3.       Run  $M$  on  $w_j$  for  $k$  steps.
4.       If it accepts:
5.           Flag = TRUE
6.       For all strings  $w$  of length  $|w_j|$ :
7.           Run  $M$  on  $w$  for  $k$  steps.
8.           If  $M$  does not halt, Flag = FALSE
9.       If Flag, *accept*.

You'll see that if an input  $\langle M \rangle$  is in  $\overline{L_3}$ , then there is some input  $w_j$  that  $M$  accepts such that  $M$  halts on every other string of the same length. For large enough  $k$ ,  $R$  will run  $M$  on this  $w_j$  for long enough for it to accept on line 3. At this point,  $R$  will run  $M$  on every other string of the same length on line 6, and, again for large enough  $k$ , each of these runs will halt. At this point,  $R$  will exit the loop from lines 6 to 8 with Flag TRUE, at which point it will accept.

On the other hand, if  $\langle M \rangle \in L_3$ , there is no input  $w_j$  that  $M$  accepts such that  $M$  will halt on every other string of the same length. In this case, when we run  $R$ , every string that  $M$  accepts will elad us to the loop from lines 6 to 8, but there will always be some string on which  $M$  does not halt in this loop. So Flag will always be set to FALSE, and so  $R$  will never accept.

**Question 4.** [10 MARKS]

Consider the language  $L_4$ :

$$L_4 = \left\{ \langle M, N \rangle \mid \begin{array}{l} \text{There exists some input } w \text{ such that} \\ M \text{ accepts } w, N \text{ loops on } w, \text{ or both} \end{array} \right\}.$$

Is  $L_4$  recognizable? Prove your answer.

**Solution:**

$L_4$  is not recognizable:

$P =$  “On input  $\langle M, w \rangle$ :

1. Let  $M_0 =$  “On input  $\langle x \rangle$ :
  1. *Reject*.”
2. Let  $N_0 =$  “On input  $\langle x \rangle$ :
  1. Run  $M$  on  $w$ .
  2. *Accept*.”
3. Return  $\langle M_0, N_0 \rangle$ .”

**Proof that**  $\langle M, w \rangle \in \overline{\text{HALT}}$  **iff**  $\langle M_0, N_0 \rangle \in L_4$ :

$\Rightarrow$ ) Suppose that  $\langle M, w \rangle \in \overline{\text{HALT}}$ .

Then, for any  $\langle x \rangle$  to  $N_0$ ,  $N_0$  loops on line 1.

So  $N_0$  always loops.

So there is some input on which  $N_0$  loops.

Therefore, there exists some  $w$  such that  $M_0$  accepts  $w$  or  $N_0$  loops on  $w$ .

So  $\langle M_0, N_0 \rangle \in L_4$ .

$\Leftarrow$ ) Suppose that  $\langle M, w \rangle \notin \overline{\text{HALT}}$ .

Then, for any  $\langle x \rangle$  to  $N_0$ ,  $N_0$  will eventually finish running line 1.

So  $N_0$  never loops on any input.

On the other hand, there also is no input that  $M_0$  will accept.

So there does not exist any  $w$  that  $M$  will accept or on which  $N$  will loop, and so  $\langle M_0, N_0 \rangle \notin L_4$ .

**Question 5.** [10 MARKS]

Consider the language  $L_4$  from the previous question:

$$L_4 = \left\{ \langle M, N \rangle \mid \begin{array}{l} \text{There exists some input } w \text{ such that} \\ M \text{ accepts } w, N \text{ loops on } w, \text{ or both} \end{array} \right\}.$$

Is  $L_4$  co-recognizable? Prove your answer.

**Solution:**

**Solution:**

$L_4$  is not co-recognizable:

$P =$  “On input  $\langle M, w \rangle$ :

1. Let  $M_0 =$  “On input  $\langle x \rangle$ :
  1. Run  $M$  on  $w$ .
  2. *Accept*.”
2. Let  $N_0 =$  “On input  $\langle x \rangle$ :
  1. *Reject*.”
3. Return  $\langle M_0, N_0 \rangle$ .”

*you'll notice we've just switched the TMs we used in the last answer!*

**Proof that**  $\langle M, w \rangle \in \text{HALT}$  **iff**  $\langle M_0, N_0 \rangle \in L_4$ :

$\Rightarrow$ ) Suppose that  $\langle M, w \rangle \in \text{HALT}$ .

Then, for any  $\langle x \rangle$  to  $M_0$ ,  $M_0$  accepts.

So there is some input that  $M_0$  accepts.

Therefore, there exists some  $w$  such that  $M_0$  accepts  $w$  or  $N_0$  loops on  $w$ .

So  $\langle M_0, N_0 \rangle \in L_4$ .

$\Leftarrow$ ) Suppose that  $\langle M, w \rangle \notin \text{HALT}$ .

Then, for any input  $\langle x \rangle$  to  $M_0$ ,  $M_0$  loops.

So  $M_0$  never accepts.

So, for every input,  $M_0$  does not accept and  $N_0$  rejects (does not loop).

So  $\langle M_0, N_0 \rangle \notin L_4$ .

**Question 6.** [10 MARKS]

Consider the language  $L_6$ :

$$L_6 = \left\{ \langle M \rangle \mid \begin{array}{l} \exists \text{ a TM } N \text{ such that for all inputs } w, \\ N \text{ loops on } w \text{ iff } M \text{ accepts } w \end{array} \right\}.$$

Prove that  $L_6$  is not recognizable.

**Solution:**

*This is a actually conceptual question: the key is to note that the language*

$$\left\{ \langle M \rangle \mid \begin{array}{l} \exists \text{ a TM } N \text{ such that for all inputs } w, \\ N \text{ loops on } w \text{ iff } M \text{ accepts } w \end{array} \right\}$$

is the same as

$$\left\{ \langle M \rangle \mid \begin{array}{l} \exists \text{ a TM } N \text{ such that for all inputs } w, \\ N \text{ loops on } w \text{ iff } M \text{ accepts } w, \text{ and accepts otherwise.} \end{array} \right\}$$

which is again the same as

$$\{ \langle M \rangle \mid \exists \text{ a TM } N \text{ such that } L(N) = \overline{L(M)}. \}$$

This  $N$  exists iff  $L(N) = \overline{L(M)}$  is recognizable, so

$$\{ \langle M \rangle \mid L(M) \text{ is both recognizable and co-recognizable.} \}$$

Which finally gives us

$$\{ \langle M \rangle \mid L(M) \text{ is decidable.} \}$$

We know from class that this language is not recognizable.

If you get this far, or use the proof from class that DECIDABLE is not recognizable, you get full marks.

If you go for a mapping reduction, use something like this:



$P =$  “On input  $\langle M, w \rangle$ :

1. Let  $M_0 =$  “On input  $\langle N', x \rangle$ :
  1. Run  $M$  on  $w$ .
  2. Run  $N'$  on  $x$ .
  3. *Accept*.”
2. Return  $\langle M_0 \rangle$ .”

**Proof that**  $\langle M, w \rangle \in \overline{\text{HALT}}$  **iff**  $\langle M_0 \rangle \in L_6$ :

$\Rightarrow$ ) Suppose that  $\langle M, w \rangle \in \overline{\text{HALT}}$ .

Then, for any  $\langle N', x \rangle$  to  $M_0$ ,  $M_0$  loops on line 1.

So  $M_0$  never accepts.

So  $L_{M_0} = \emptyset$ .

Therefore, a TM  $N$  that accepts every input will loop iff  $M_0$  accepts.

So  $\langle M_0 \rangle \in L_6$ .

$\Leftarrow$ ) Suppose that  $\langle M, w \rangle \notin \overline{\text{HALT}}$ .

Then, for any  $\langle N', x \rangle$  to  $M_0$ ,  $M_0$  will eventually finish running line 1.

So  $M_0$  accepts iff line 2 finishes.

So  $L_{M_0} = \{ \langle N', x \rangle \mid N' \text{ halts on input } x \} = \text{HALT}$ .

Now, if there were a TM that would loop iff  $M_0$  accepts, we could build a TM  $N$  that would loop iff  $M_0$  accepts, and would accept otherwise.

The language of this  $N$  would be  $\overline{\text{HALT}}$ . But this is impossible, since  $\overline{\text{HALT}}$  is not recognizable.

So no such  $N$  exists, and  $\langle M_0 \rangle \notin L_6$ .